

The Top-Degree Mass of Unit-Distance Graphs and the Price of Node Privacy

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Abstract

Erdős asked how many pairs among n points in the plane can be at distance exactly 1. We study an unbalanced version of this question and show that it determines the cost of a basic task in data privacy. Let $\Lambda_k(n)$ be the largest possible sum of the k largest degrees in the unit-distance graph of at most n points. We prove that the smallest expected error with which the number of unit-distance pairs can be released under ε -node differential privacy is $\Theta(\Lambda_{\lceil 1/\varepsilon \rceil}(n))$, with absolute constants, for every $\varepsilon \in (0, 1]$. We also show that $\Lambda_k(n)$ equals, up to a factor of 2, the maximum number of incidences between n points and k unit circles. We then compare norms on the plane. $\Lambda_k(n) = \Theta(kn)$ if the unit circle contains a segment; $\Lambda_k(n) = O(n^{2/3}k^{2/3} + n)$ if the norm is strictly convex, and a norm found by Valtr attains this bound for every k ; for almost all norms, $\Lambda_k(n) = \Theta(n)$ up to a logarithmic factor. For the Euclidean norm the optimal error is $\Theta(n)$ when $\varepsilon \geq n^{-1/2}$, a factor $1/\varepsilon$ below the standard Laplace mechanism. It is polynomially larger than for almost all norms if and only if Erdős's unit-distance conjecture fails, which was shown in 2026. Between these ranges the Euclidean question is an open problem on unbalanced unit-circle incidences.

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1 Introduction

Let P be a finite set of points in the plane and let $G(P)$ be its *unit-distance graph*: two points are adjacent if their distance is exactly 1. Erdős [13] asked for the maximum number $u(n)$ of edges of $G(P)$ over n -point sets. The upper bound $u(n) = O(n^{4/3})$ of Spencer, Szemerédi and Trotter [26] has not been improved. Erdős conjectured $u(n) = n^{1+o(1)}$; in 2026 this conjecture was disproved [20, 1], and Sawin [24] gave sets with more than $n^{1.014}$ unit distances for arbitrarily large n .

This paper studies a refinement of $u(n)$ that measures how unevenly unit distances can be distributed among the points. For a finite set P and an integer $k \geq 1$, let $S_k(P)$ be the sum of the k largest degrees of $G(P)$ (padded with zeros if $k > |P|$), and define the *top-degree mass*

$$\Lambda_k(n) = \max\{S_k(P) : |P| \leq n\}.$$

Thus $\Lambda_1(n) = n - 1$ (a point together with $n - 1$ points on its unit circle), and $\Lambda_n(n) = 2u(n)$. In between, $\Lambda_k(n)$ asks how many unit distances k points can see.

The privacy problem. Our motivation comes from differential privacy. Suppose each point of P is the location of one person, and we wish to publish the number $U(P)$ of unit-distance pairs. *Node differential privacy* requires that adding or removing one person, together with all pairs that contain this person, changes the distribution of the published value by at most a factor e^ε (see Section 3 for a self-contained introduction). One point can form a unit-distance pair with all others, so the standard Laplace mechanism must add noise of order n/ε . We



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ask for the *minimax* error

$$\text{err}^*(\varepsilon, n) = \inf_M \sup_{|P| \leq n} \mathbb{E} |M(P) - U(P)|,$$

where M ranges over ε -node-private algorithms and n is a public bound on the number of points. Our first result says that, up to constant factors, this privacy question is the geometric question above.

► **Theorem 1** (Reduction). *For every norm on $\mathbb{R}^2$, every $n \geq 1$ and every $\varepsilon \in (0, 1]$, with $k = \lceil 1/\varepsilon \rceil$,*

$$\frac{1}{8(1+e)} \Lambda_k(n) \leq \text{err}^*(\varepsilon, n) \leq 2 \Lambda_k(n).$$

The upper bound is attained by an algorithm that is ε -node-private on all finite point sets. The same holds, with Λ_k defined accordingly, for every family of point sets closed under deleting points and every graph that is determined by pairs of points (for instance, any distance-bin graph).

The ingredients are known in the privacy literature: a bounded-degree surrogate for the edge count due to Kasiviswanathan, Nissim, Raskhodnikova and Smith [19], and a lower bound by deleting and re-inserting high-degree vertices, which is closely related to the down-sensitivity analysis of Dong, Fang, Yi, Tao and Machanavajjhala [9]. We do not claim these tools as new. What Theorem 1 adds is that, over a family closed under deletion, the worst-case cost is given by a single extremal function. This turns the privacy question into a question about point sets, which is the subject of the rest of the paper.

Geometry of the top-degree mass. The k largest degrees of $G(P)$ count incidences between P and the unit circles centred at k points of P . Let $I(n, k)$ be the maximum number of incidences between at most n points and k distinct unit circles (with arbitrary centres). Equivalently, $I(n, k)$ is the maximum number of unit distances between a set of n points and a set of k points.

► **Theorem 2** (Top-degree mass and incidences). *For every norm on $\mathbb{R}^2$ and all integers $n \geq 2$ and $1 \leq k \leq n$,*

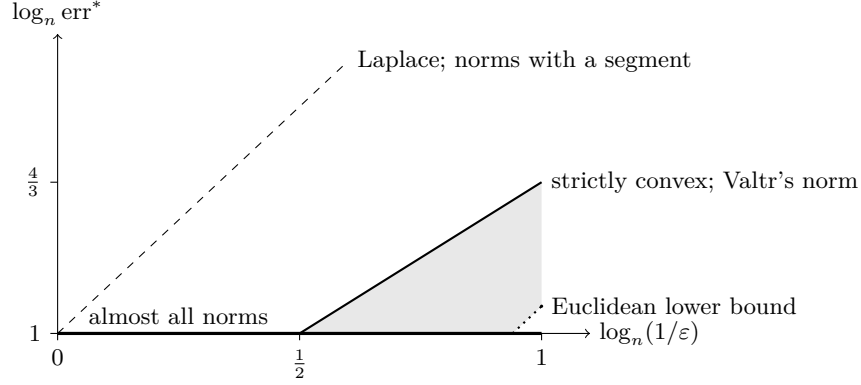
$$\max \left\{ n - 1, \frac{2k u(n)}{n} \right\} \leq \Lambda_k(n) \leq I(n, k),$$

where $u(n)$ is the maximum number of unit distances among n points in this norm. If $k \leq n/2$, then also $I(n, k) \leq 2\Lambda_k(n)$.

Which values can Λ_k take? We answer this across all norms on $\mathbb{R}^2$. A norm is *strictly convex* if its unit circle contains no segment. Valtr [30] found a strictly convex norm in which n points can span $\Omega(n^{4/3})$ unit distances; we show that his norm is extremal for every k , not only for $k = n$. Alon, Bucić and Sauermaun [2] proved that for almost all norms (in the sense of Baire category) every n points span at most $n \log_2 n$ unit distances.

► **Theorem 3** (Top-degree mass across norms). *There are absolute constants $C, c > 0$ such that for all $n \geq 8$ and $1 \leq k \leq n$:*

1. *If the unit circle contains a segment, then $\lfloor n/2 \rfloor \min\{k, n-1\} \leq \Lambda_k(n) \leq (n-1)k$.*
2. *If the norm is strictly convex, then $\Lambda_k(n) \leq C(n^{2/3}k^{2/3} + n)$.*
3. *For Valtr's norm $\|(x, y)\| = |y| + \sqrt{x^2 + y^2}$, we have $\Lambda_k(n) \geq c(n^{2/3}k^{2/3} + n)$.*



■ **Figure 1** Exponent of the optimal error against $\log_n(1/\varepsilon)$. Shaded: the range allowed for the Euclidean norm by Theorem 5; its lower edge is drawn with $\delta = 0.06$, while the best known value is $\delta \geq 0.014$.

82 **4.** For almost all norms, $n - 1 \leq \Lambda_k(n) \leq 2n \log_2 n$.

83 Since every norm is either strictly convex or has a segment on its unit circle, Theorems 1
84 and 3 describe the worst possible privacy cost over all norms.

85 ► **Corollary 4** (Privacy cost across norms). For $n \geq 8$ and $1/(n-1) \leq \varepsilon \leq 1$, the minimax
86 error $\text{err}^*(\varepsilon, n)$ is $\Theta(n/\varepsilon)$ if the unit circle contains a segment, so the Laplace mechanism is
87 optimal; it is $O(n + n^{2/3}\varepsilon^{-2/3})$ for every strictly convex norm, with equality up to constants
88 for Valtr's norm; and it lies between $\Omega(n)$ and $O(n \log n)$ for almost all norms.

89 **The Euclidean plane.** The Euclidean norm is strictly convex, so its privacy cost lies below
90 Valtr's curve. The averaging bound in Theorem 2 brings in $u(n)$.

91 ► **Theorem 5** (Euclidean plane). There are absolute constants $C, c > 0$ such that for the
92 Euclidean norm, all $n \geq 8$ and all $\varepsilon \in [1/n, 1]$,

$$93 \quad c \left(n + \frac{u(n)}{\varepsilon n} \right) \leq \text{err}^*(\varepsilon, n) \leq C \left(n + n^{2/3} \varepsilon^{-2/3} \right).$$

94 In particular $\text{err}^*(\varepsilon, n) = \Theta(n)$ for $\varepsilon \geq n^{-1/2}$. Moreover, the following are equivalent:

- 95 1. $u(n) \geq n^{1+\delta}$ for some $\delta > 0$ and infinitely many n ;
- 96 2. there are $\delta' > 0$ and privacy levels $\varepsilon_n \in (0, 1]$ such that $\text{err}^*(\varepsilon_n, n) \geq n^{1+\delta'}$ for infinitely
97 many n .

98 By the 2026 results, both hold: for every $\eta \in [0, 0.014)$ there are infinitely many n with
99 $\text{err}^*(n^{-1+\eta}, n) \geq cn^{1.014-\eta}$, while for almost all norms the error is $O(n \log n)$ at every ε .

100 Figure 1 shows the resulting picture. For the Euclidean norm, the region between the two
101 bounds of Theorem 5 is open. By Theorem 2, closing it is the same as determining $I(n, k)$
102 for unit circles when $\sqrt{n} \leq k \leq n$, an unbalanced version of the unit-distance problem. At
103 $k = n$ this is the unit-distance problem itself; Theorem 3(3) shows that any improvement
104 over the incidence bound must use properties of the Euclidean norm that Valtr's norm does
105 not have. We state the question in Section 8.

106 **Summary and provenance.** Table 1 lists the results and says which parts are new. We
107 regard the geometric results (Theorems 2, 3, and 5) as the main contribution. Theorem 1 is
108 the bridge; Section 7 adds an algorithm that does not need to know Λ_k , at a logarithmic cost.

Result	Statement	Status
Theorem 1	$\text{err}^*(\varepsilon, n) = \Theta(\Lambda_{\lceil 1/\varepsilon \rceil}(n))$	known tools; new form
Theorem 2	$\Lambda_k(n)$ and $I(n, k)$ agree up to a factor 2	new (short)
Theorem 3(1),(4)	segment norms; almost all norms	new (short); uses [2]
Theorem 3(2)	strictly convex norms	standard incidence bound
Theorem 3(3)	Valtr's norm is extremal for every k	new; extends [30]
Theorem 5	Euclidean window; separation iff $u(n) \geq n^{1+\Omega(1)}$	new connection
Theorem 12	algorithm without knowledge of Λ_k	known tools [19, 23]

■ **Table 1** Results and their provenance.

Related work. *Unit distances and incidences.* Besides the bounds above, the unit-distance problem is surveyed by Brass, Moser and Pach [5]. The $O(n^{4/3})$ bound follows from incidence bounds for points and curves [29, 7, 21], and Székely's crossing-number argument [28] gives it for any family of curves in which two curves meet in at most two points and two points lie on at most two curves. For normed planes, Swanepoel's survey [27] describes the known results, including Valtr's construction. After the disproof of Erdős's conjecture [20, 1], Sawin [24] gave the exponent 1.014 and Emmerich [12] optimised explicit finite certificates. We use these results only through Theorem 2; we do not use their structure.

Node privacy. Node differential privacy for graphs was studied by Blocki, Blum, Datta and Sheffet [4] and by Kasiviswanathan, Nissim, Raskhodnikova and Smith [19], who introduced the bounded-degree flow statistic we use. For a family closed under deletion, the restricted sensitivity of Blocki, Blum, Datta and Sheffet is $\Lambda_1(n)$ in our notation, which leads to error about $\Lambda_1(n)/\varepsilon$. Theorem 1 replaces this by $\Lambda_{\lceil 1/\varepsilon \rceil}(n)$, which is never larger; for strictly convex norms it is smaller by a factor of order at least $\varepsilon^{-2/3}$. Raskhodnikova and Smith [23] developed Lipschitz extensions and the generalized exponential mechanism used in Section 7. Instance-dependent guarantees in terms of down sensitivity appear in work of Asi and Duchi [3], Fang, Dong and Yi [14], and Dong, Fang, Yi, Tao and Machanavajjhala [9]; see also Cummings and Durfee [8]. Recent node-private results include counting connected components [18], continual observation [17] and graphon estimation [6]. Raskhodnikova, Smith, Wagaman and Zavalov [22] study node privacy in the *local* model, where each person randomizes their own edge list; we work in the central model, where a trusted curator holds the data. We are not aware of earlier work that links node privacy to incidence geometry.

2 Overview of the proofs

Why the privacy cost is $\Lambda_{\lceil 1/\varepsilon \rceil}$. The upper bound uses one public threshold. The surrogate g_D of [19] changes by at most D when a point is added or removed, and it undercounts $U(P)$ by at most the degree tail $\text{tail}_D(P) = \sum_p (\deg_P(p) - D)_+$. Set $k = \lceil 1/\varepsilon \rceil$ and $D = \Lambda_k(n)/k$. The vertices of degree above D form a set S , and $\text{tail}_D(P) \leq S_{|S|}(P) - |S|D \leq \Lambda_{|S|}(n) - |S|D$. The average $\Lambda_j(n)/j$ cannot increase with j , so this is at most $\Lambda_k(n)$ for every value of $|S|$. Laplace noise of scale $D/\varepsilon \leq \Lambda_k(n)$ completes the bound.

For the lower bound, take a set P^* with $S_k(P^*) = \Lambda_k(n)$, where now $k \approx 1/\varepsilon$, and remove its k points of largest degree. Re-inserting them one at a time is a path of $k \leq 1/\varepsilon$ neighbouring inputs along which the count grows by at least $\Lambda_k(n)/2$. A private algorithm cannot distinguish the two ends of such a path well, so it errs by $\Omega(\Lambda_k(n))$ on one of them.

142 **Why Λ_k is an incidence count.** If C is the set of k points of largest degree in P , then
 143 $S_k(P)$ is the number of incidences between P and the unit circles centred at points of C .
 144 Conversely, given n points and k unit circles, adding the k centres to the point set turns
 145 incidences into degrees. Splitting the points into two halves keeps the total size below n at a
 146 loss of a factor 2.

147 **Why the norms differ.** If the unit circle contains a segment, two parallel rows of points give
 148 a complete bipartite unit-distance graph, so $\Lambda_k(n) = \Theta(kn)$. If the norm is strictly convex,
 149 two distinct unit circles meet in at most two points and two points lie on at most two unit
 150 circles; the crossing-number argument then gives $I(n, k) = O(n^{2/3}k^{2/3} + n + k)$. For Valtr's
 151 norm, the upper half of each unit circle is a parabolic arc, and the map $(x, y) \mapsto (x, y + x^2/2)$
 152 sends all these arcs to segments of lines. The standard grid construction for point-line
 153 incidences, at the right aspect ratio, then gives $I(n, k) = \Omega(n^{2/3}k^{2/3})$ for every $\sqrt{n} \leq k \leq n$.
 154 For almost all norms the bound of [2] on $u(n)$ caps $\Lambda_k(n) \leq 2u(n)$.

155 **Why the 2026 result enters.** The averaging bound $\Lambda_k(n) \geq 2k u(n)/n$ holds because the k
 156 largest of n degrees have at least the average degree. With $k = \lceil 1/\varepsilon \rceil$ close to n , Theorem 1
 157 turns a superlinear $u(n)$ into a superlinear privacy cost. Conversely $\Lambda_k(n) \leq \Lambda_n(n) = 2u(n)$,
 158 so no other source of superlinear cost exists.

159 **3 Preliminaries**

160 **Geometry.** Fix a norm $\|\cdot\|$ on $\mathbb{R}^2$ with unit circle $\mathbb{S} = \{x : \|x\| = 1\}$; a *unit circle* is a
 161 translate $c + \mathbb{S}$. For a finite set P , $\deg_P(p)$ is the number of $q \in P$ with $\|p - q\| = 1$, and
 162 $U(P)$ is the number of edges of $G(P)$. For a finite family Γ of curves, $I(P, \Gamma)$ is the number
 163 of pairs $(p, \gamma) \in P \times \Gamma$ with $p \in \gamma$. Since $\mathbb{S} = -\mathbb{S}$, a point p lies on $c + \mathbb{S}$ exactly when
 164 $\|p - c\| = 1$. We write $\mathcal{P}_n$ for the family of point sets with at most n points. The functions
 165 $\Lambda_k(n)$ and $I(n, k)$ are non-decreasing in both arguments, and $\Lambda_k(n) = \Lambda_n(n) = 2u(n)$ for
 166 $k \geq n$.

167 **Node differential privacy, for geometers.** An *algorithm* M maps each finite point set P to
 168 a random real number $M(P)$. Two point sets are *node neighbours* if one is obtained from the
 169 other by adding one point. The word “node” refers to the graph $G(P)$: adding a point adds
 170 a vertex together with all its edges.

171 **► Definition 6.** Let $\varepsilon > 0$ and let $\mathcal{F}$ be a family of finite point sets. An *algorithm* M
 172 is ε -node-private on $\mathcal{F}$ if for all node neighbours $P, Q \in \mathcal{F}$ and every measurable $T \subseteq \mathbb{R}$,
 173 $\Pr[M(P) \in T] \leq e^\varepsilon \Pr[M(Q) \in T]$.

174 The definition says that the output reveals little about whether any single person is
 175 present. Smaller ε means stronger privacy; the range $\varepsilon \leq 1$ is the usual one. We use three
 176 standard facts [10, 11]. (i) *Laplace mechanism*: if $|f(P) - f(Q)| \leq \Delta$ for all node neighbours,
 177 then $f(P) + Z$ with Z drawn from the Laplace density $\frac{\varepsilon}{2\Delta} e^{-\varepsilon|z|/\Delta}$ is ε -node-private, and
 178 $\mathbb{E}|Z| = \Delta/\varepsilon$. For $f = U$ one has $\Delta = n - 1$, which gives error $(n - 1)/\varepsilon$; we call this
 179 the *standard Laplace mechanism*. (ii) *Groups*: if P and Q are joined by a path of h node
 180 neighbours in $\mathcal{F}$, then $\Pr[M(P) \in T] \leq e^{h\varepsilon} \Pr[M(Q) \in T]$. (iii) The following two-point
 181 bound, which we prove for completeness.

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182 ► **Lemma 7.** *Let M be ε -node-private on $\mathcal{F}$, and let $P, Q \in \mathcal{F}$ be joined by a path of h*
 183 *node neighbours in $\mathcal{F}$ with $h\varepsilon \leq 1$. If $U(Q) - U(P) = \Delta > 0$, then $\mathbb{E}|M(P) - U(P)|$ or*
 184 *$\mathbb{E}|M(Q) - U(Q)|$ is at least $\Delta/(2(1+e))$.*

185 **Proof.** Let α be the larger of the two expected errors and $T = [U(P) + \Delta/2, \infty)$. By
 186 Markov's inequality $\Pr[M(P) \in T] \leq 2\alpha/\Delta$ and $\Pr[M(Q) \notin T] \leq 2\alpha/\Delta$. By (ii), $1 - 2\alpha/\Delta \leq$
 187 $\Pr[M(Q) \in T] \leq e \cdot 2\alpha/\Delta$, so $\alpha \geq \Delta/(2(1+e))$. ◀

4 The reduction

189 ► **Lemma 8.** *For every n , the sequence $k \mapsto \Lambda_k(n)/k$ is non-increasing. Hence $\Lambda_j(n) \leq$*
 190 *$(j/k)\Lambda_k(n)$ for $j \geq k$.*

191 **Proof.** For a fixed P , $S_k(P)/k$ is the average of the k largest degrees (with zero padding),
 192 which does not increase when k grows. A pointwise maximum of non-increasing sequences is
 193 non-increasing. ◀

194 For a graph G and a real number $D \geq 0$, let

$$195 \quad g_D(G) = \max \left\{ \sum_{e \in E(G)} x_e : 0 \leq x_e \leq 1, \sum_{e \ni v} x_e \leq D \text{ for every vertex } v \right\}.$$

196 This is the value of the flow statistic of [19], written as a fractional degree-constrained
 197 subgraph (Appendix A). We write $g_D(P)$ for $g_D(G(P))$.

198 ► **Lemma 9.** *For every real $D \geq 0$: (a) if P, Q are node neighbours then $|g_D(P) - g_D(Q)| \leq D$;*
 199 *(b) $U(P) - \text{tail}_D(P) \leq g_D(P) \leq U(P)$ for every P .*

200 **Proof.** (a) Let $Q = P \cup \{q\}$. An optimal solution for Q , restricted to the edges not containing
 201 q , is feasible for P and loses at most $\sum_{e \ni q} x_e \leq D$. An optimal solution for P , extended
 202 by zeros, is feasible for Q . (b) The upper bound holds since $x_e \leq 1$. For the lower bound
 203 put $x_e = \min\{1, D/\deg u, D/\deg v\}$ for $e = uv$. At each vertex v the sum is at most
 204 $\deg v \cdot \min\{1, D/\deg v\} \leq D$. Each edge loses $1 - x_e \leq (1 - D/\deg u)_+ + (1 - D/\deg v)_+$,
 205 and summing over edges gives $\sum_v \deg v (1 - D/\deg v)_+ = \text{tail}_D(P)$. ◀

206 ► **Lemma 10.** *For every $P \in \mathcal{P}_n$ and real $D \geq 0$, $\text{tail}_D(P) \leq \max_{0 \leq j \leq n} (\Lambda_j(n) - jD)$, where*
 207 *$\Lambda_0 = 0$.*

208 **Proof.** Let S be the set of points of degree larger than D and $j = |S|$. Then $\text{tail}_D(P) =$
 209 $\sum_{p \in S} \deg_P(p) - jD \leq S_j(P) - jD \leq \Lambda_j(n) - jD$. ◀

210 The proof below uses only properties (a) and (b) of Lemma 9. Any statistic with these
 211 two properties can replace g_D ; for integer D this includes the integral statistic of Appendix A.

212 ► **Theorem 1 (Reduction).** *For every norm on $\mathbb{R}^2$, every $n \geq 1$ and every $\varepsilon \in (0, 1]$, with*
 213 *$k = \lceil 1/\varepsilon \rceil$,*

$$214 \quad \frac{1}{8(1+e)} \Lambda_k(n) \leq \text{err}^*(\varepsilon, n) \leq 2 \Lambda_k(n).$$

215 *The upper bound is attained by an algorithm that is ε -node-private on all finite point sets. The*
 216 *same holds, with Λ_k defined accordingly, for every family of point sets closed under deleting*
 217 *points and every graph that is determined by pairs of points (for instance, any distance-bin*
 218 *graph).*

Proof. *Upper bound.* Let $k = \lceil 1/\varepsilon \rceil$, $\lambda = \Lambda_k(n)$ and $D = \lambda/k$. The algorithm outputs $g_D(P) + Z$ with Z Laplace of scale D/ε ; it is ε -node-private on all finite sets by Lemma 9(a). For $j \geq k$, Lemma 8 gives $\Lambda_j(n) - jD \leq 0$, and for $j < k$ we have $\Lambda_j(n) - jD \leq \lambda$. By Lemma 10 and Lemma 9(b), for every $P \in \mathcal{P}_n$,

$$\mathbb{E}|g_D(P) + Z - U(P)| \leq \text{tail}_D(P) + \mathbb{E}|Z| \leq \lambda + \frac{\lambda}{k\varepsilon} \leq 2\lambda.$$

Lower bound. Let $m = \min\{\lceil 1/\varepsilon \rceil, n\} \geq 1$ and choose $P^* \in \mathcal{P}_n$ with $S_m(P^*) = \Lambda_m(n)$. Let C consist of $\min\{m, |P^*|\}$ points of P^* of largest degree and $P_0 = P^* \setminus C$. Adding the points of C to P_0 one at a time gives a path of $|C| \leq m$ node neighbours inside $\mathcal{P}_n$. Every edge counted in $\sum_{c \in C} \deg(c) = \Lambda_m(n)$ is counted at most twice, so $U(P^*) - U(P_0) \geq \Lambda_m(n)/2$. Since $m\varepsilon \leq 1$, Lemma 7 shows that every ε -node-private algorithm has expected error at least $\Lambda_m(n)/(4(1+e))$ on P_0 or on P^* . Finally, if $m = n$ then $\Lambda_k(n) = \Lambda_m(n)$; otherwise $k \leq \lceil 1/\varepsilon \rceil + 1 \leq 2m$, and Lemma 8 gives $\Lambda_k(n) \leq 2\Lambda_m(n)$.

Other families. The upper bound used only Lemma 9, which holds for all graphs. The lower bound used only that P_0 and the intermediate sets belong to the family. Both hold for any family closed under deleting points and any graph in which adjacency of two points depends only on these two points. ◀

► **Remark 11.** The algorithm needs the value $\Lambda_k(n)$. If only an upper bound $\lambda \geq \Lambda_k(n)$ is known, the same proof with $D = \lambda/k$ gives error at most 2λ . For the Euclidean norm this yields error $O(n + n^{2/3}\varepsilon^{-2/3})$ from Theorem 3(2). Theorem 12 below removes the need for any bound at a logarithmic cost.

5 Top-degree mass and incidences

► **Theorem 2** (Top-degree mass and incidences). *For every norm on $\mathbb{R}^2$ and all integers $n \geq 2$ and $1 \leq k \leq n$,*

$$\max\left\{n-1, \frac{2k u(n)}{n}\right\} \leq \Lambda_k(n) \leq I(n, k),$$

where $u(n)$ is the maximum number of unit distances among n points in this norm. If $k \leq n/2$, then also $I(n, k) \leq 2\Lambda_k(n)$.

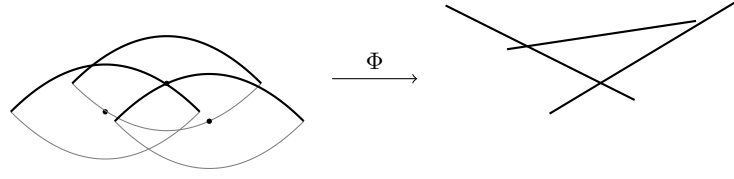
Proof. *Lower bounds.* Placing $n-1$ points on the unit circle around a further point gives $S_1 \geq n-1$. If $|P| = n$ and $U(P) = u(n)$, the average degree is $2u(n)/n$, and the k largest degrees have at least this average, so $S_k(P) \geq 2k u(n)/n$.

$\Lambda_k(n) \leq I(n, k)$. Let $P \in \mathcal{P}_n$ and let C be a set of $\min\{k, |P|\}$ points of P of largest degree. The unit circles $c + \mathbb{S}$, $c \in C$, are distinct, and $S_k(P) = \sum_{c \in C} |P \cap (c + \mathbb{S})| = I(P, \{c + \mathbb{S}\}_{c \in C}) \leq I(n, k)$.

$I(n, k) \leq 2\Lambda_k(n)$ for $k \leq n/2$. Let B be a set of at most n points, R a set of k centres, and $\Gamma_R = \{r + \mathbb{S}\}_{r \in R}$. Split B into two parts of at most $\lceil n/2 \rceil$ points each, and let B' be a part with $I(B', \Gamma_R) \geq I(B, \Gamma_R)/2$. The set $Q = B' \cup R$ has at most $\lceil n/2 \rceil + k \leq n$ points. Each $r \in R$ has $\deg_Q(r) \geq |B' \cap (r + \mathbb{S})|$, hence $\Lambda_k(n) \geq S_k(Q) \geq \sum_{r \in R} \deg_Q(r) \geq I(B, \Gamma_R)/2$. ◀

Lemma 10 also bounds the degree tail of every set by the top-degree mass. With Theorem 3(2) this gives, for every strictly convex norm,

$$\text{tail}_D(P) \leq \max_j (C(n^{2/3}j^{2/3} + n) - jD) = O(n^2/D^2 + n) \quad (P \in \mathcal{P}_n, D \geq 1). \quad (1)$$



■ **Figure 2** Left: three unit circles of Valtr's norm, each bounded by two parabolic arcs. Right: the map $\Phi(x, y) = (x, y + x^2/2)$ sends the upper arcs (bold) to line segments whose slope is the first coordinate of the centre. Point sets with many point-line incidences therefore yield many unit distances.

Both terms are needed. A star with $2D$ leaves, repeated $\lfloor n/(2D+1) \rfloor$ times, has tail $\Omega(n)$. Since $\text{tail}_D(P) \geq S_k(P) - kD$ for every k , Theorem 3(3) with $k \asymp n^2/D^3$ shows that for Valtr's norm the first term is attained when $n^{1/3} \lesssim D \lesssim n^{1/2}$. We note that (1) applies the incidence bound once, to the circles centred at the high-degree points; summing a bound on the number of j -rich points over all $j > D$ would lose a logarithmic factor.

6 Norms on the plane

► **Theorem 3** (Top-degree mass across norms). *There are absolute constants $C, c > 0$ such that for all $n \geq 8$ and $1 \leq k \leq n$:*

1. *If the unit circle contains a segment, then $\lfloor n/2 \rfloor \min\{k, n-1\} \leq \Lambda_k(n) \leq (n-1)k$.*
2. *If the norm is strictly convex, then $\Lambda_k(n) \leq C(n^{2/3}k^{2/3} + n)$.*
3. *For Valtr's norm $\|(x, y)\| = |y| + \sqrt{x^2 + y^2}$, we have $\Lambda_k(n) \geq c(n^{2/3}k^{2/3} + n)$.*
4. *For almost all norms, $n-1 \leq \Lambda_k(n) \leq 2n \log_2 n$.*

Proof of (1). Let the segment $\{v + tw : |t| \leq 1\}$ lie on $\mathbb{S}$. Put $m = \lfloor n/2 \rfloor$, $a_i = (i/m)w$ and $b_j = v + (j/m)w$ for $0 \leq i, j < m$. Then $b_j - a_i = v + \frac{j-i}{m}w$ lies on the segment, so all m^2 pairs $\{a_i, b_j\}$ are at unit distance. (The points are distinct, since a segment on $\mathbb{S}$ does not lie on a line through the origin.) Every point has degree at least m , and $2m \geq n-1$, so $S_k \geq m \min\{k, n-1\}$. The upper bound holds because every degree is at most $n-1$. ◀

Proof of (2). In a strictly convex norm, two distinct unit circles meet in at most two points, and two distinct points lie on at most two common unit circles (Lemma 17 in Appendix C). For such families, Székely's crossing-number argument [28], or the theorem of Pach and Sharir [21], gives $I(P, \Gamma) \leq C_0(|P|^{2/3}|\Gamma|^{2/3} + |P| + |\Gamma|)$; we include the argument in Appendix C. By Theorem 2, $\Lambda_k(n) \leq I(n, k) \leq C_0(n^{2/3}k^{2/3} + n + k)$, and $k \leq n$. ◀

Proof of (3). The norm $\|(x, y)\| = |y| + \sqrt{x^2 + y^2}$ is the sum of the Euclidean norm and a seminorm, so it is strictly convex [30, 27]. For a centre $c = (a, b)$ and a point $p = (x, y)$ with $y \geq b$, the equation $\|p - c\| = 1$ is equivalent to

$$y = b + \frac{1}{2}(1 - (x - a)^2), \quad |x - a| \leq 1.$$

The bijection $\Phi(x, y) = (x, y + x^2/2)$ (Figure 2) maps this arc into the line $Y = aX + (b + \frac{1}{2} - \frac{a^2}{2})$. Conversely, every point (X, Y) with $|X - a| \leq 1$ on the line $Y = aX + \beta$ is the image of a point at unit distance from the centre $(a, \beta - \frac{1}{2} + \frac{a^2}{2})$.

Now use the grid construction for point-line incidences. For integers $A, B \geq 1$, the $2A^2B$ points (i, j) with $1 \leq i \leq A$, $1 \leq j \leq 2AB$ and the AB^2 lines $j = si + t$ with $1 \leq s \leq B$, $1 \leq t \leq AB$ have A^2B^2 incidences, since each line contains A of the points.

290 Apply $(i, j) \mapsto (\frac{i}{2A}, \frac{j}{4AB})$: the lines get slopes $a = \frac{s}{2B} \in (0, \frac{1}{2}]$, and all points have $X \in (0, \frac{1}{2}]$,
 291 so $|X - a| \leq 1$. Applying Φ^{-1} to the points and taking the centres described above yields
 292 $2A^2B$ points and AB^2 distinct centres with at least A^2B^2 pairs at unit distance. Choosing A
 293 and B for given n and k (Appendix C) gives $I(\lfloor n/2 \rfloor, k) \geq n^{2/3}k^{2/3}/53$ for $\sqrt{n} \leq k \leq \lfloor n/2 \rfloor$.
 294 By Theorem 2 and monotonicity this gives (3), with $\Lambda_k(n) \geq n - 1$ for $k < \sqrt{n}$. ◀

295 **Proof of (4).** Alon, Bucić and Sauermaun [2] prove that for all norms on $\mathbb{R}^2$ outside a
 296 meagre set, every n points span at most $n \log_2 n$ unit distances. Then $\Lambda_k(n) \leq \Lambda_n(n) =$
 297 $2u(n) \leq 2n \log_2 n$. The lower bound $n - 1$ holds in every norm. ◀

298 **Proof of Corollary 4.** Let $k = \lceil 1/\varepsilon \rceil \leq n - 1$ and apply Theorem 1. With a segment,
 299 $\Lambda_k(n) \geq \lfloor n/2 \rfloor / \varepsilon$, and the standard Laplace mechanism has error $(n - 1)/\varepsilon$. For strictly
 300 convex norms, (2) gives $2\Lambda_k(n) = O(n + n^{2/3}k^{2/3})$ and $k \leq 2/\varepsilon$; (3) gives the matching lower
 301 bound for Valtr's norm. For almost all norms, (4) gives $\Omega(n)$ and $O(n \log n)$. ◀

302 ▶ **Theorem 5 (Euclidean plane).** *There are absolute constants $C, c > 0$ such that for the*
 303 *Euclidean norm, all $n \geq 8$ and all $\varepsilon \in [1/n, 1]$,*

$$304 \quad c \left(n + \frac{u(n)}{\varepsilon n} \right) \leq \text{err}^*(\varepsilon, n) \leq C (n + n^{2/3} \varepsilon^{-2/3}).$$

305 *In particular $\text{err}^*(\varepsilon, n) = \Theta(n)$ for $\varepsilon \geq n^{-1/2}$. Moreover, the following are equivalent:*

- 306 1. $u(n) \geq n^{1+\delta}$ for some $\delta > 0$ and infinitely many n ;
- 307 2. there are $\delta' > 0$ and privacy levels $\varepsilon_n \in (0, 1]$ such that $\text{err}^*(\varepsilon_n, n) \geq n^{1+\delta'}$ for infinitely
 308 many n .

309 *By the 2026 results, both hold: for every $\eta \in [0, 0.014)$ there are infinitely many n with*
 310 *$\text{err}^*(n^{-1+\eta}, n) \geq cn^{1.014-\eta}$, while for almost all norms the error is $O(n \log n)$ at every ε .*

311 **Proof.** Let $k = \lceil 1/\varepsilon \rceil$, so $1/\varepsilon \leq k \leq \min\{n, 2/\varepsilon\}$. By Theorems 1 and 2, $\text{err}^*(\varepsilon, n) \geq$
 312 $\frac{1}{8(1+\varepsilon)} \max\{n - 1, 2u(n)/(\varepsilon n)\}$. The upper bound follows from Theorems 1 and 3(2). For
 313 $\varepsilon \geq n^{-1/2}$ we have $n^{2/3}\varepsilon^{-2/3} \leq n$.

314 For the equivalence, if $u(n) \geq n^{1+\delta}$ then at $\varepsilon = 1/n$ the lower bound is at least $cu(n) \geq$
 315 $cn^{1+\delta}$, which exceeds $n^{1+\delta/2}$ for large n . Conversely, $\text{err}^*(\varepsilon, n) \leq 2\Lambda_k(n) \leq 2\Lambda_n(n) = 4u(n)$
 316 for every ε , so a lower bound $n^{1+\delta'}$ on the error forces $u(n) \geq n^{1+\delta'}/4$. Finally, Sawin [24]
 317 gives $u(n) > n^{1.014}$ for infinitely many n ; for these n and $\varepsilon = n^{-1+\eta}$ the lower bound is at
 318 least $cn^{-\eta}u(n) \geq cn^{1.014-\eta}$. The comparison with almost all norms is Corollary 4. ◀

319 **What the separation does and does not say.** The separation in Theorem 5 appears only
 320 for $\varepsilon \leq n^{-1+0.014}$, that is, for very strong privacy. At such ε the optimal error is already
 321 comparable to the count itself, so we do not see it as a recommendation for practice. Its
 322 content is structural: the shape of the Euclidean rate curve near $\varepsilon = 1/n$ is decided exactly
 323 by whether $u(n)$ is polynomially superlinear, and the 2026 results settle this. For larger ε
 324 the curve is decided by $I(n, k)$ with $k < n$, where the gap in Theorem 5 is open.

325 7 Algorithms and other distance graphs

326 **An algorithm that does not need Λ_k .** The algorithm of Theorem 1 uses the unknown
 327 number $\Lambda_k(n)$. The following result, which combines the surrogate of [19] with the generalized
 328 exponential mechanism of Raskhodnikova and Smith [23], adapts to the input instead. It is
 329 not new as a method; we include it because its guarantee has a direct geometric reading.

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330 ► **Theorem 12.** *Let $K = 1 + \lceil \log_2 n \rceil$. There is an ε -node-private algorithm such that for*
 331 *every $P \in \mathcal{P}_n$ and $\beta \in (0, \frac{1}{2})$, with probability at least $1 - \beta$, its error is $O(S_t(P) + t)$ with*
 332 *$t = \lceil \log(K/\beta)/\varepsilon \rceil$. Consequently the error is $O(\log(K/\beta) (\Lambda_{\lceil 1/\varepsilon \rceil}(n) + 1/\varepsilon))$.*

333 The algorithm selects a threshold D from $\{1, 2, 4, \dots, 2^{K-1}\}$ privately and then releases
 334 $g_D(P)$ with Laplace noise; the proof is in Appendix B. The bound $S_t(P)$ is the number of
 335 incidences between P and the t richest unit circles centred in P . For instance, if every point
 336 of P has degree at most Δ , the error is $O(\Delta \log(K/\beta)/\varepsilon)$ regardless of n .

337 **Surrogates.** Theorem 1 needs real thresholds, because $\Lambda_k(n)/k$ need not be an integer; this
 338 is one reason to use the fractional statistic g_D . The integral statistic $f_D(G)$, the maximum
 339 number of edges of a subgraph of maximum degree at most D , satisfies the same two
 340 properties for integer D and can be computed by b -matching [25, 15]. With D rounded up
 341 it gives Theorem 1 up to an additive $1/\varepsilon$ (Appendix A). Both statistics are computable in
 342 polynomial time from $G(P)$.

343 **Other distance graphs.** Theorem 1 applies to every family of point sets that is closed
 344 under deleting points. Two examples are worked out in Appendix E. If the points lie on
 345 a grid $\frac{1}{q}\mathbb{Z}^2$ and the target distance r satisfies $qr \in \mathbb{Z}$, every degree is at most the number
 346 Δ of representations of $(qr)^2$ as a sum of two squares, and the error is $\Theta(\Delta/\varepsilon)$ whenever
 347 $n \geq \lceil 1/\varepsilon \rceil (\Delta + 1)$. If the points are δ -separated and adjacency means a distance in $[r - t, r + t]$,
 348 every degree is $O((t + \delta)(r + t + \delta)/\delta^2)$; in this case releasing g_D for D equal to this bound
 349 is private on all inputs and exact on separated inputs. Finally, Appendix D treats a class of
 350 point sets defined by an upper bound on the number of j -rich points for every j ; this class
 351 is not closed under deletion, so Theorem 1 does not apply, and we give separate matching
 352 bounds.

353 8 Open problems

354 **The Euclidean window.** For $\alpha \in [\frac{1}{2}, 1]$ let $\gamma(\alpha) = \limsup_{n \rightarrow \infty} \log \Lambda_{\lceil n^\alpha \rceil}(n) / \log n$ for the
 355 Euclidean norm, and let $1 + \delta^* = \gamma(1)$ be the unit-distance exponent. Theorems 2 and 3 give

$$356 \quad \max\{1, \alpha + \delta^*\} \leq \gamma(\alpha) \leq \frac{2+2\alpha}{3},$$

357 with $\delta^* \geq 0.014$, and for Valtr's norm the upper bound is attained. Is the lower bound the
 358 truth? In words: can a few unit circles be much richer than average while the unit-distance
 359 graph as a whole stays sparse? By Theorem 2 this is a question about incidences between n
 360 points and k unit circles. We are not aware of results on this unbalanced version beyond the
 361 bounds above.

362 **Instance-wise bounds.** Theorem 1 concerns the worst case over $\mathcal{P}_n$. A local version would
 363 compare an algorithm, at each input P , with the best algorithm on a small neighbourhood
 364 of P . Deleting points is controlled by $S_t(P)$, as in Theorem 12; adding points brings in the
 365 external quantity $\max_{c \notin P} |P \cap (c + \mathbb{S})|$ and its analogues for several inserted points.

366 **Higher dimensions.** In $\mathbb{R}^3$, $u(n)$ lies between $n^{4/3} \log \log n$ [5] and $n^{295/197+o(1)}$ [31], so the
 367 window is wider and both ends are open. In $\mathbb{R}^d$ with $d \geq 4$, points on two orthogonal circles
 368 of radius $1/\sqrt{2}$ form a complete bipartite unit-distance graph, as in Theorem 3(1), and the
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Appendix

A Surrogate statistics

The flow statistic. Kasiviswanathan, Nissim, Raskhodnikova and Smith [19] define the following network for a graph $G = (V, E)$ and a capacity $D \geq 0$. Take a source, a sink, and two copies v_ℓ, v_r of each vertex. Add arcs of capacity D from the source to each v_ℓ and from each v_r to the sink, and for each edge uv add arcs (u_ℓ, v_r) and (v_ℓ, u_r) of capacity 1. Let $\phi_D(G)$ be the maximum flow value. The definition makes sense for real D .

► **Proposition 13.** For every graph G and real $D \geq 0$, $\phi_D(G)/2 = g_D(G)$.

Proof. Given a feasible x for g_D , send x_{uv} units along both arcs of each edge. This is a feasible flow of value $2 \sum_e x_e$. Conversely, from a feasible flow y set $x_{uv} = (y_{u_\ell v_r} + y_{v_\ell u_r})/2$. Then $0 \leq x_e \leq 1$, and $\sum_{e \ni v} x_e$ is half the flow through v_ℓ plus half the flow through v_r , which is at most D . The objective is half the flow value. ◀

Hence Lemma 9 is a property of the statistic of [19]; our only additions are the explicit fractional solution in part (b), which gives the bias bound for every real D , and the remark that real thresholds are needed in Theorem 1.

The integral statistic. For an integer $D \geq 0$ let $f_D(G)$ be the maximum number of edges in a subgraph of G with maximum degree at most D .

► **Proposition 14.** For integer $D \geq 0$: (a) $|f_D(G) - f_D(G - v)| \leq D$; (b) $|E| - \text{tail}_D(G) \leq f_D(G) \leq g_D(G) \leq |E|$, and both inner inequalities can be strict; (c) f_D and g_D can be computed in polynomial time.

Proof. (a) An optimal subgraph for G has at most D edges at v ; removing them gives a feasible subgraph of $G - v$. Every feasible subgraph of $G - v$ is feasible for G . (b) Repeatedly delete an edge at a vertex of current degree above D ; each deletion lowers $\sum_v (\deg v - D)_+$ by at least one, so at most $\text{tail}_D(G)$ edges are deleted. Integral solutions are feasible for g_D . For a triangle and $D = 1$, $f_1 = 1 < 3/2 = g_1$; for a star with more than D leaves, $g_D = D < |E|$. (c) g_D is a maximum flow; f_D is a maximum-cardinality simple b -matching with $b \equiv D$ [25, 15]. ◀

With $D' = \lceil \Lambda_k(n)/k \rceil$ in place of D in the proof of Theorem 1, the bias is at most $\text{tail}_{D'} \leq \text{tail}_D \leq \Lambda_k(n)$ and the noise is at most $(D + 1)/\varepsilon \leq \Lambda_k(n) + 1/\varepsilon$, so $f_{D'}$ gives error at most $2\Lambda_k(n) + 1/\varepsilon$.

B The adaptive algorithm

Let $\mathcal{D} = \{1, 2, 4, \dots, 2^{K-1}\}$ with $K = 1 + \lceil \log_2 n \rceil$, so $\max \mathcal{D} \geq n - 1$. Fix $\beta \in (0, \frac{1}{2})$, split $\varepsilon = \varepsilon_1 + \varepsilon_2$ with $\varepsilon_1 = \varepsilon_2 = \varepsilon/2$, and let $a = \log(2/\beta)$.

Algorithm. Run the generalized exponential mechanism of Raskhodnikova and Smith [23, Theorem 1.4] with privacy parameter ε_1 on the scores $q_D(P) = -g_D(P) + aD/\varepsilon_2$, $D \in \mathcal{D}$, where score q_D is declared to have sensitivity D ; let $\hat{D}$ be the output. Release $g_{\hat{D}}(P) + Z$ with Z Laplace of scale $\hat{D}/\varepsilon_2$.

► **Lemma 15.** The algorithm is ε -node-private. With probability at least $1 - \beta$ its error is at most $\min_{D \in \mathcal{D}} \{\text{tail}_D(P) + aD/\varepsilon_2 + 4D \log(2K/\beta)/\varepsilon_1\}$.

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Proof. Each score q_D has sensitivity at most D by Lemma 9(a), since aD/ε_2 does not depend on P . The selection step is ε_1 -node-private by [23, Theorem 1.4], the release step is ε_2 -node-private given $\hat{D}$, and the composition is ε -node-private. By [23, Theorem 1.4], with probability at least $1 - \beta/2$, $q_{\hat{D}}(P) \leq \min_D \{q_D(P) + 4D \log(2K/\beta)/\varepsilon_1\}$. With probability at least $1 - \beta/2$, $|Z| \leq a\hat{D}/\varepsilon_2$. On both events,

$$|g_{\hat{D}}(P) + Z - U(P)| \leq U(P) - g_{\hat{D}}(P) + a\hat{D}/\varepsilon_2 = U(P) + q_{\hat{D}}(P),$$

and $U(P) + q_D(P) \leq \text{tail}_D(P) + aD/\varepsilon_2$ by Lemma 9(b). $\blacktriangleleft$

► **Lemma 16.** *Let $d_{(1)} \geq d_{(2)} \geq \dots$ be the degrees of $G(P)$, padded with zeros. For every integer $t \geq 1$, $\min_{D \geq 0} \{\text{tail}_D(P) + tD\} = S_t(P)$, and $\min_{D \in \mathcal{D}} \{\text{tail}_D(P) + tD\} \leq 2S_t(P) + t$.*

Proof. For every $D \geq 0$, $\text{tail}_D(P) + tD \geq \sum_{i \leq t} (d_{(i)} - D) + tD = S_t(P)$, with equality at $D = d_{(t+1)}$. For the grid, if $d_{(t+1)} = 0$ take $D = 1$: only the top t points have positive degree, so $\text{tail}_1(P) \leq S_t(P)$. Otherwise take $D \in \mathcal{D}$ with $d_{(t+1)} \leq D \leq 2d_{(t+1)}$; then $\text{tail}_D(P) + tD \leq S_t(P) - t d_{(t+1)} + 2t d_{(t+1)} \leq 2S_t(P)$. $\blacktriangleleft$

Proof of Theorem 12. By Lemma 15, the error is at most $\min_{D \in \mathcal{D}} \{\text{tail}_D(P) + t'D\}$ with $t' = 2a/\varepsilon + 8 \log(2K/\beta)/\varepsilon = O(t)$. By Lemma 16 and since $S_{t'} \leq \lceil t'/t \rceil S_t$, this is $O(S_t(P) + t)$. For the second statement let $k = \lceil 1/\varepsilon \rceil$. Then $S_t(P) \leq \Lambda_{\max\{t,k\}}(n) \leq (t/k + 1)\Lambda_k(n)$ by Lemma 8, and $t/k \leq \log(K/\beta) + 1$. $\blacktriangleleft$

C Deferred geometric proofs

► **Lemma 17.** *Let $\mathbb{S}$ be the unit circle of a strictly convex norm. Two distinct translates of $\mathbb{S}$ meet in at most two points, and two distinct points lie on at most two common translates of $\mathbb{S}$.*

Proof. Let K be the unit ball and $v \neq 0$. Choose coordinates (s, τ) with τ measured along v . Each line $\{s = s_0\}$ that meets the interior of K meets K in a segment $[\tau_-(s_0), \tau_+(s_0)]$, where τ_- is convex and τ_+ is concave; at the two extreme values of s the intersection is a single point, by strict convexity. The boundary of $K + v$ on the same line is $\{\tau_-(s_0) + |v|, \tau_+(s_0) + |v|\}$. Since $\tau_- \leq \tau_+$ and $|v| > 0$, a common boundary point must satisfy $\tau_+(s_0) = \tau_-(s_0) + |v|$, that is, $h(s_0) = |v|$ for $h = \tau_+ - \tau_-$. Strict convexity of K means that the graphs of $\tau_{\pm}$ contain no segment, so h is strictly concave, and it vanishes at the extreme values of s . A strictly concave function takes a positive value at most twice, which proves the first claim. For the second, $p, q \in c + \mathbb{S}$ means $c \in (p + \mathbb{S}) \cap (q + \mathbb{S})$ because $\mathbb{S} = -\mathbb{S}$, and these two translates are distinct. $\blacktriangleleft$

► **Lemma 18.** *There is an absolute constant C_0 such that for every strictly convex norm, every finite set P and every finite family Γ of distinct unit circles, $I(P, \Gamma) \leq C_0(|P|^{2/3}|\Gamma|^{2/3} + |P| + |\Gamma|)$.*

Proof. This is the argument of Székely [28]. Let $N = |P|$ and $m = |\Gamma|$. Draw a multigraph H on P : for every $\gamma \in \Gamma$ containing $j \geq 2$ points of P , join consecutive points along γ by the j arcs between them. Then H has at least $I(P, \Gamma) - m$ edges. By Lemma 17, two points are joined by arcs of at most two circles, and each circle contributes at most two such arcs, so every edge has multiplicity at most 4. Two arcs of the same circle do not cross, and two circles meet in at most two points, so the drawing has at most m^2 crossings. The crossing lemma for multigraphs with bounded multiplicity [28] states that a multigraph

with N vertices, e edges and multiplicity at most μ has crossing number at least $c_1 e^3 / (\mu N^2)$ whenever $e \geq c_2 \mu N$, for absolute constants $c_1, c_2 > 0$. Hence either $I(P, \Gamma) - m < 4c_2 N$ or $m^2 \geq c_1 (I(P, \Gamma) - m)^3 / (4N^2)$, and in both cases the claim follows. ◀

Choice of parameters in Theorem 3(3). Let $n \geq 8$, $m = \lfloor n/2 \rfloor \geq 4$ and $\sqrt{n} \leq k \leq m$. Put $A_0 = (m/2)^{2/3} k^{-1/3}$ and $B_0 = k^{2/3} (m/2)^{-1/3}$, so that $2A_0^2 B_0 = m$ and $A_0 B_0^2 = k$. Since $k \leq m$ and $m \geq 4$ we have $(m/2)^2 \geq k$, so $A_0 \geq 1$; since $k^2 \geq n \geq m/2$ we have $B_0 \geq 1$. Let $A = \lfloor A_0 \rfloor \geq A_0/2$ and $B = \lfloor B_0 \rfloor \geq B_0/2$. The construction in the proof of Theorem 3(3) uses at most m points and at most k centres and has at least $A^2 B^2 \geq A_0^2 B_0^2 / 16 = (mk/2)^{2/3} / 16$ incidences. Since $m \geq n/3$, this is at least $(nk/6)^{2/3} / 16 \geq n^{2/3} k^{2/3} / 53$. By Theorem 2 (the set Q in its proof has at most $m + k \leq n$ points), $\Lambda_k(n) \geq n^{2/3} k^{2/3} / 53$. For $m < k \leq n$, monotonicity gives $\Lambda_k(n) \geq \Lambda_m(n) \geq n^{2/3} m^{2/3} / 53 \geq n^{2/3} k^{2/3} / 111$, using $m \geq n/3 \geq k/3$ and $m \geq \sqrt{n}$. For $k < \sqrt{n}$, $\Lambda_k(n) \geq n - 1 \geq n/2 \geq n^{2/3} k^{2/3} / 2$. Hence (3) holds with $c = 1/222$. ◀

D A class defined by rich points

In this appendix the norm is Euclidean. For $j \geq 1$ let $R_j(P)$ be the number of points of P of degree at least j . For $C \geq 1$ and $D_0 \geq 1$, call P (C, D_0) -regular if $R_j(P) \leq C|P|^2/j^3$ for every integer $j > D_0$. Deleting a point can violate this condition, so the class is not closed under deletion, and Theorem 1 does not apply. The bound $R_j(P) = O(n^2/j^3 + n/j)$ holds for every set by the incidence bound, so the condition removes exactly the term n/j coming from many stars of degree about j .

► **Proposition 19.** *If P is (C, D_0) -regular, $|P| = n$, and $D \geq D_0$, then $\text{tail}_D(P) = \sum_{j>D} R_j(P) = O(Cn^2/D^2)$. Hence the algorithm $g_D(P) + \text{Lap}(D/\varepsilon)$ with $D \asymp (C\varepsilon n^2)^{1/3} \geq D_0$ has expected error $O(C^{1/3} n^{2/3} \varepsilon^{-2/3})$, and the algorithm of Theorem 12 has error $O(C^{1/3} n^{2/3} (\log(K/\beta)/\varepsilon)^{2/3})$ with probability $1 - \beta$ whenever $D_0 \leq (Cn^2\varepsilon/\log(K/\beta))^{1/3}$.*

Proof. $\text{tail}_D(P) = \sum_{j>D} R_j(P) \leq Cn^2 \sum_{j>D} j^{-3} = O(Cn^2/D^2)$. The first algorithm is private on all inputs by Lemma 9(a), and its error is $O(D/\varepsilon + Cn^2/D^2)$. For the second, Lemma 15 bounds the error by $O(\min_{D \in \mathcal{D}, D \geq D_0} \{Cn^2/D^2 + D \log(K/\beta)/\varepsilon\})$; optimise over D . ◀

► **Proposition 20.** *There are absolute constants $a, b > 0$ and n_0 such that for $n \geq n_0$, $1 \leq C \leq n$, $D_0 \geq 1$ and $aC^{1/2} n^{-1/2} \leq \varepsilon \leq 1$, every ε -node-private algorithm has expected error at least $bC^{1/3} n^{2/3} \varepsilon^{-2/3}$ on some (C, D_0) -regular set of at most n points.*

Proof. Let $h = \max\{1, \lfloor 1/(4\varepsilon) \rfloor\}$, so $1/(8\varepsilon) \leq h \leq 1/\varepsilon$. Choose a large and $\alpha \leq \frac{1}{2}$ small with $2\alpha a^{-2/3} \leq \frac{1}{2}$, and set $L = \lfloor \alpha(Cn^2/h)^{1/3} \rfloor$. Since $(Cn^2/h)^{1/3} \geq (Cn^2\varepsilon)^{1/3} \geq a^{1/3} C^{1/2} n^{1/2}$, for $n \geq n_0$ we get $L \geq \frac{\alpha}{2} (Cn^2/h)^{1/3} \geq 2$ and $h(L+1) \leq 2\alpha C^{1/3} n^{2/3} \varepsilon^{-2/3} \leq n/2$. Take h far-apart gadgets, each consisting of a potential centre c_i and L points on its unit circle, with angles chosen so that no two of these L points are at unit distance, together with $n - h(L+1)$ isolated padding points. Let P_i contain all non-centre points and $c_1, \dots, c_i$. Then $|P_h| = n$, $|P_i| \geq n/2$, consecutive sets are node neighbours, and $U(P_h) - U(P_0) = hL$. For $2 \leq j \leq L$ the points of degree at least j are the inserted centres, so $R_j(P_i) \leq h \leq C|P_i|^2/L^3 \leq C|P_i|^2/j^3$ by the choice of α , and $R_j(P_i) = 0$ for $j > L$. Thus every P_i is (C, D_0) -regular. Since $h\varepsilon \leq 1$, Lemma 7 gives error at least $hL/(2(1+e)) = \Omega(C^{1/3} n^{2/3} h^{2/3}) = \Omega(C^{1/3} n^{2/3} \varepsilon^{-2/3})$. ◀

The class is useful as a conditional statement: it describes inputs on which the rich points are spread over many degree levels. We do not know a natural family of point sets with

many unit distances that satisfies the condition and is not built from stars, and we leave this as a question.

E Grid points and separated sets

Grid points. Let $P \subset \frac{1}{q}\mathbb{Z}^2$ and let the target distance r satisfy $qr \in \mathbb{Z}$. Points of $\frac{1}{q}\mathbb{Z}^2$ at distance r from a grid point correspond to integer solutions of $x^2 + y^2 = (qr)^2$, so every degree is at most $\Delta = r_2((qr)^2)$, the number of such representations, and $\Delta \leq (qr)^{O(1/\log \log(qr))}$ [16]. Grid point sets form a family closed under deletion, with $\Lambda_k \leq k\Delta$. A lattice point together with the Δ grid points on its circle shows $\Lambda_k \geq k\Delta$ when $n \geq k(\Delta + 1)$ (use k far-apart copies). By Theorem 1 the error is $\Theta(\Delta/\varepsilon)$ when $n \geq \lceil 1/\varepsilon \rceil(\Delta + 1)$; the Laplace mechanism applied to U itself achieves the upper bound. If arbitrary inputs are first rounded to a fixed public grid, neighbours are mapped to equal or neighbouring grid sets, so the rounded statistic inherits privacy.

Separated sets and annuli. Fix $\delta > 0$, $r > 0$, $t \geq 0$, and let two points be adjacent when their distance lies in $[r - t, r + t]$. Consider the family of δ -separated sets, in which every two points have distance at least δ ; it is closed under deletion. The disks of radius $\delta/2$ around the neighbours of a point p are disjoint and lie in the annulus around p with radii $\max\{0, r - t - \delta/2\}$ and $r + t + \delta/2$. If the inner radius is positive, this annulus has area $2\pi r(2t + \delta)$; otherwise $r \leq t + \delta/2$ and the area is at most $\pi(2t + \delta)^2$. In both cases the area is at most $\pi(2t + \delta)(2r + 2t + \delta)$, so every degree is at most

$$\Delta_{\text{sep}} = \frac{4(2t + \delta)(2r + 2t + \delta)}{\delta^2} = O\left(\frac{rt + t^2}{\delta^2} + \frac{r + t}{\delta} + 1\right).$$

For $t \leq r$ this is $O(rt/\delta^2 + r/\delta + 1)$. For $t > r$ the term t^2/δ^2 cannot be dropped: then every point within distance $r + t$ of p is a neighbour of p , and a disk of radius $t \geq \delta$ contains $\Omega(t^2/\delta^2)$ points of a δ -separated grid. The algorithm $g_D(P) + \text{Lap}(D/\varepsilon)$ with $D = \Delta_{\text{sep}}$ is ε -node-private on *all* finite sets by Lemma 9(a); on δ -separated sets every degree is at most D , so $g_D(P) = U(P)$ by Lemma 9(b) and the only error is the noise, $\Delta_{\text{sep}}/\varepsilon$ in expectation. The same argument applies to threshold graphs (distance at most r), with degree $O(r^2/\delta^2 + 1)$.

Several distance bins. For a histogram of distances with bins $b_1, \dots, b_L$, adding a point changes all bins at once. A simple private release splits $\varepsilon = \varepsilon_1 + \dots + \varepsilon_L$ and runs the one-bin algorithm with parameter ε_i in bin i .